

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 35%

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
Weightage	25%	25%	25%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

**Constraint-Based Problem
Solving Assessment Rubric**

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Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving (6 Marks) (BL4–BL5) – Reframed with New Scenarios

Q1. Dataset: Hospital Patient Records

PID	Symptoms Observed
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P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "**Fever**". Identify the valid association rules and justify why they satisfy the given constraints. (BL4 – 1.2 Marks)

Q2. Dataset: Online Learning Platform

Student ID	Courses Enrolled
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S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining. (BL4 – 1.2 Marks)

Q3. Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
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T1	Data Mining, Python Programming
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Transaction ID	Books Borrowed
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation. (BL5 – 1.2 Marks)

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the **"Computer Science"** level. Explain how the course hierarchy affects the discovery of association rules. (BL5 – 1.2 Marks)

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies. (BL4 – 1.2 Marks)

ANNEXURE A – CONSTRAINT-BASED PROBLEM SOLVING

Q1. Hospital Patient Records

Minimum support = 50% = 3 out of 5 patients.

Item	Count	Support
Fever	4	80%
Cough	3	60%
Headache	3	60%
Fatigue	1	20%

Frequent single items: {Fever}, {Cough}, {Headache}.

For a valid association rule, the combined itemset must have at least 50% support. The pairs containing Fever are: {Fever, Cough} = $2/5 = 40\%$, {Fever, Headache} = $2/5 = 40\%$, and {Fever, Cough, Headache} = $1/5 = 20\%$.

Answer: No association rule satisfies all the conditions: minimum support 50%, minimum confidence 70%, and the requirement that every generated rule must include **Fever**. The constraint removes all rules that do not contain Fever, and the remaining Fever-based combinations fail the minimum support requirement.

Q2. Online Learning Platform

Minimum support = 40% = 2 out of 5 students. The constraint requires every itemset to contain at least two courses.

Itemset	Students	Support	Frequent?
{Python, SQL}	S1, S4	40%	Yes
{Python, Data Science}	S2, S4	40%	Yes
{Python, Machine Learning}	S5	20%	No
{SQL, Data Science}	S4	20%	No
{SQL, Machine Learning}	S3	20%	No
{Data Science, Machine Learning}	None	0%	No
{Python, SQL, Data Science}	S4	20%	No

Frequent itemsets satisfying the constraint: {Python, SQL} and {Python, Data Science}.

Pruning explanation: Single-course itemsets are not considered as final valid results because they violate the “at least two courses” constraint. Candidate itemsets with support below 40% are also removed, so they do not need to be expanded further. This reduces the number of candidates checked during mining.

Q3. Library Borrowing Records

Constraint: Maximum itemset size = 3. Since no minimum-support value is specified, the itemsets below are shown with their observed support counts (all itemsets appearing in at least one transaction).

Itemset	Count	Support
{Data Mining}	4	80%
{Python Programming}	3	60%

Itemset	Count	Support
{Database Systems}	3	60%
{Artificial Intelligence}	1	20%
{Data Mining, Python Programming}	2	40%
{Data Mining, Database Systems}	2	40%
{Python Programming, Database Systems}	2	40%
{Data Mining, Artificial Intelligence}	1	20%
{Data Mining, Python Programming, Database Systems}	1	20%

Frequent itemsets: The itemsets considered are all 1-item, 2-item and 3-item sets listed above, subject to any minimum-support threshold chosen by the instructor.

Complexity reduction: The Apriori algorithm stops generating candidates larger than 3 items. Therefore, 4-item, 5-item and larger combinations are never generated or counted, reducing memory use and computation.

Q4. University Course Hierarchy

The hierarchy can be viewed as **Engineering** → **Computer Science** → **Machine Learning**, while Information Technology is another branch under Engineering.

Constraint: Generate rules only at the **Computer Science** level.

Pattern / Rule	Count	Support / Confidence
Computer Science	3 students	60% support
Computer Science → Machine Learning	2 students	Support = 40%, Confidence = $2/3 = 66.7\%$
Machine Learning → Computer Science	2 students	Support = 40%, Confidence = 100%

Explanation: The hierarchy determines the level at which items are analyzed. By restricting mining to the Computer Science level, rules involving unrelated branches such as Information Technology are excluded. This gives more focused rules and reduces the search space.

Q5. Insurance Customer Records

Constraint: Age Group = “Young”. Only C1, C3 and C5 are used.

Young Customers	Income Level	Policy Purchased
C1	High	Health Insurance
C3	Low	Travel Insurance
C5	Medium	Vehicle Insurance

Meaningful patterns:

1. **Young** $\wedge$ **High Income** → **Health Insurance** (1 occurrence among 3 Young customers; confidence within this exact subgroup = 100%).
2. **Young** $\wedge$ **Low Income** → **Travel Insurance** (1 occurrence; confidence = 100%).
3. **Young** $\wedge$ **Medium Income** → **Vehicle Insurance** (1 occurrence; confidence = 100%).

Marketing use: These patterns can support targeted campaigns. For example, young customers with high income can be targeted for Health Insurance, while low-income young customers may be targeted for Travel Insurance and medium-income young customers for Vehicle Insurance. The Age Group constraint